C A N A R Y

Do companies committing accounting fraud write their annual reports differently?

A pre-registered novelty study on SEC 10-K filings.

Trained: five small autoencoders in PyTorch, one per fraud case (each tested against industry peers).

Reused: MiniLM, a pretrained sentence encoder. Used as-is, no fine-tuning.

Tested: six historical accounting frauds; methodology locked in git before any number was looked at.

Arsenii Chan · CSC 44800 Artificial Intelligence · Spring 2026 · CCNY

scan to try

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If an autoencoder trains only on clean annual reports from the same industry and year, will it rank known fraud filings as more unusual than its peers?

The one-line answer this report defends:

At N = 6 frauds, with strict pre-registration: no useful autoencoder signal in four of five test cases; Enron couldn't be evaluated under the frozen rule; the one case the autoencoder did rank highly (Lehman) is n = 1, and a 1990s TF-IDF baseline beats the autoencoder there.

The contract is the contribution.

git-tagged validation-spec-frozen, committed before any validation script ran.

Out-of-sample, always

Leave-one-cohort-out (LOCO): for each held-out fraud, the autoencoder is trained on clean filings from other cohorts only. The fraud and its peers are never in the training set.

Time-controlled training

Within the LOCO training set, only filings dated on or before that fraud's filing date are eligible. The autoencoder's training corpus cannot include post-fraud disclosure language. (The MiniLM encoder isn't time-controlled.)

Single pass, no tuning

Architecture, hyperparameters, sentence cap, seeds: all committed and tagged before validation ran. scripts/06_validate.py runs once. No model tuning after validation.

Whatever the test produces, the contract makes the result interpretable.

```
git tag: validation-spec-frozen
at 98d2585  ·  2026-05-01 13:42:42 EDT
```

Only Lehman ranks better than random expectation; even there, a 1990s baseline beats the autoencoder.

Cohort	Rank	Note
Lehman (LEH, FY 2007) ⚠	**3 / 13**	*n = 1, baseline beats autoencoder*
HealthSouth (HRC)	7 / 12	at or below random
Valeant (VRX)	8 / 13	at or below random
Tyco (TYC)	9 / 13	at or below random
WorldCom (WCOM)	12 / 13	at or below random
Enron (ENE, FY 2000)	—	*not evaluable: filed earliest, no eligible training data*

Aggregate hit-rates

hit@1: **0.00** vs random 0.08

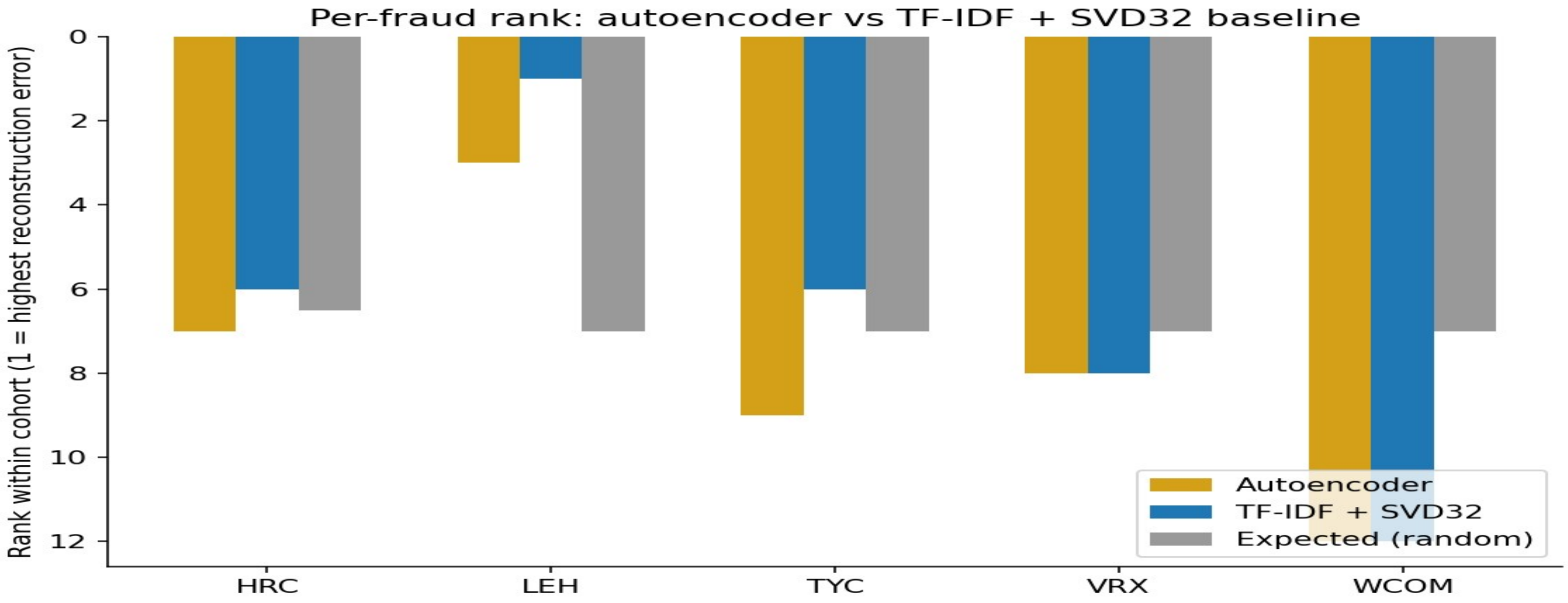
hit@3: **0.20** vs random 0.23

hit@5: **0.20** vs random 0.39

⚠ Lehman is the only rank better than random expectation, and a 1990s baseline beats the autoencoder there too (slide 5). The aggregate is entirely Lehman.

A 1990s baseline matched or beat the autoencoder on every cohort.

Post-hoc: tests whether the autoencoder earned its complexity, does not rescue the primary result.



How to read the chart: shorter bar = lower rank number = better detection (rank 1 is most anomalous). Lehman: autoencoder rank 3/13, TF-IDF rank 1/13. Aggregate hit@5 is 0.40 (TF-IDF) vs 0.20 (autoencoder). The pre-registered model does not beat the trivial baseline.

QUESTIONS I EXPECT

Q **Did you check a trivial baseline?**

Yes. TF-IDF + truncated-SVD, same bottleneck dimension, same training corpora. It matches or beats the autoencoder on every cohort. The honest claim is that the autoencoder adds nothing here. Slide 5.

Q **The baseline is post-hoc. Isn't that HARKing (hypothesizing after results)?**

The pre-registered hypothesis was about the autoencoder, and the autoencoder result is reported unchanged. The TF-IDF baseline was added after seeing the null and is reported as post-hoc. The primary null does not depend on it.

Q **Could MiniLM have memorized 'Lehman' / 'Repo 105' / etc.?**

Tested post-hoc. Masked 93 such mentions with [ENTITY] and re-scored. Lehman's rank held at 3/13 (delta = 0). Exact-token name leakage did not explain the rank; topical leakage remains open.

CONCLUSION

What survived: pre-registration.
What failed: the autoencoder beyond the TF-IDF baseline.

A result that loses to a baseline is not evidence for the neural model. The contract held; the model did not.

What I'd do differently

Confirm every held-out fraud is evaluable under the pre-registered rule BEFORE freezing the spec. Enron taught that the hard way.

Next experiment

Same autoencoder, but compare each company against its own past filings (not industry peers). Does Lehman's MD&A look anomalous against Lehman's own prior-year MD&A?

Scan to try the pipeline

Pipeline demo (not a fraud scanner): canary-psi.vercel.app/scan

Code: github.com/ArseniiChan/canary · Tag: validation-spec-frozen

Thanks.